Practice workbook · ASCII and Unicode character sets

How characters are encoded: ASCII bits and size, finding a code by counting on from a given code, benefits of ASCII, and why Unicode was needed.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-09

1. Question 2 · part a.iii · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a.iii only.

2 Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

(2)

(ii) Identify the number of characters that can be represented using standard ASCII.

(1)

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

(2)

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

(2)

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P 7 2 5 3 8 A 0 4 2 0

1. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

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2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

2 Computers use binary digits to represent data.

(a) Sign and magnitude and two's complement are two ways of representing signed integer numbers.

(i) Convert the denary number -94 to a binary pattern using sign and magnitude representation.

(2)

(ii) 1000 1001 is a signed integer that uses two's complement representation.

Convert this signed integer to denary.

(2)

(b) Denary numbers use base 10

Identify the base for hexadecimal numbers.

(1)

- ☒ **A** 32
- ☒ **B** 16
- ☒ **C** 8
- ☒ **D** 2

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2. Question 2 · part c.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.i only.

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(c) Computers encode characters using ASCII and Unicode.

(i) State the number of characters that can be represented using standard ASCII.

(1)

(ii) The ASCII system uses 7 bits to represent a character. The ASCII code for the character 'A' using denary is 65; other alphabetical characters follow on from this in sequence.

Identify the ASCII code for 'H'.

(1)

- ☒ **A** 1000111
- ☒ **B** 1001010
- ☒ **C** 1001000
- ☒ **D** 1000101

(iii) Explain why Unicode was developed.

(2)

(Total for Question 2 = 9 marks)

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2. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.

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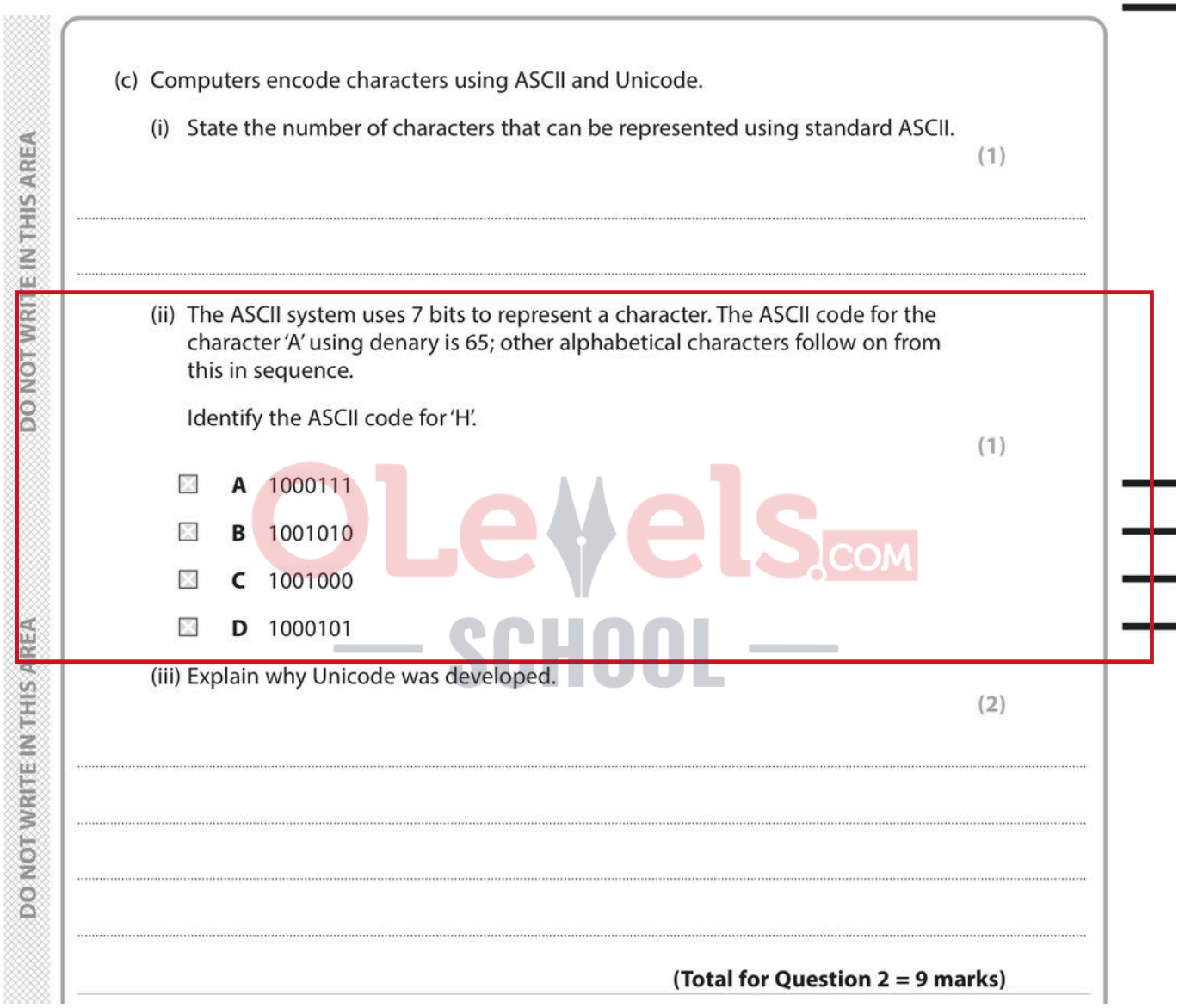
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3. Question 2 · part c.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part c.ii only.



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3. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.iii · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c.iii only.

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2020 NOV · Paper 01

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.....

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4. Question 2 · part c.iii · 2 marks · Working and answer

Working and answer

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5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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5. Question 1 · part f · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

.....

(f) Give the number of bits per character used by standard ASCII.

(1)

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

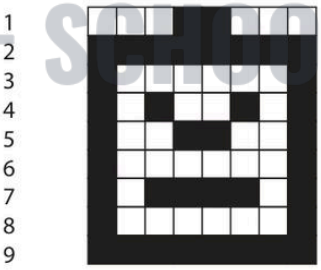


Figure 1

(i) State what is meant by the term **pixel**.

(1)

.....

(ii) Construct an expression to show the number of pixels in the image.

(1)

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5. Question 1 · part f · 1 mark · Working and answer

Working and answer

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6. Question 2 · part a.ii · 1 mark

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Further practice · Attempt part a.ii only.

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6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

3 Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

(1)

.....

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

(3)

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

(1)

☒ **A** 54

☒ **B** 191

☒ **C** 202

☒ **D** –74

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7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

(2)

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

(1)

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

(2)

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7. Question 3 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

(c) The ASCII code for the character **1** is 011 0001

(i) Give the ASCII code for the character **5**

(1)

.....

(ii) Give **two** benefits of using ASCII to encode characters.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 13 marks)

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Working and answer

8. Question 3 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

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